

Darukaa.Earth:

AI Biodiversity Intelligence Chatbot

Hackathon solution document — an AI environmental scientist, not a chatbot.

GitHub repository: <https://github.com/Palakagarwal28/Darukaa.earth-hackathon>

The repository holds the complete project: backend, knowledge base, frontend, tests, CI. Runs fully offline — no deployment, credentials, or API keys required.

1. Objective

Build an AI-powered conversational system that maintains a structured knowledge base of biodiversity and environmental metrics, understands queries about ecosystems, land and climate, generates actionable non-obvious recommendations, and supports each with scientific reasoning and evidence.

2. Tech stack

Layer	Technology	Why
API backend	FastAPI (Python) + Uvicorn	Typed endpoints, OpenAPI docs at /docs
Retrieval (RAG)	BM25 + agro synonyms (default); TF-IDF + dense retriever offline	Efficient, explainable
Knowledge store	Versioned JSON, 32 evidence cards + SQLite	Auditable; migratable to pgvector/Chroma
Reasoning	Deterministic multi-metric engine	No hallucinated advice, reproducible
Conversation	Slot-filling + SQLite memory	Clarifying Qs, context awareness
Frontend	HTML/CSS/JS, zero build step	3 tabs served by the same server
Optional	Open-Meteo (keyless) + LLM rephrase	Graceful offline fallback
Quality	pytest (12 tests) + GitHub Actions	Green on every push

3. System architecture

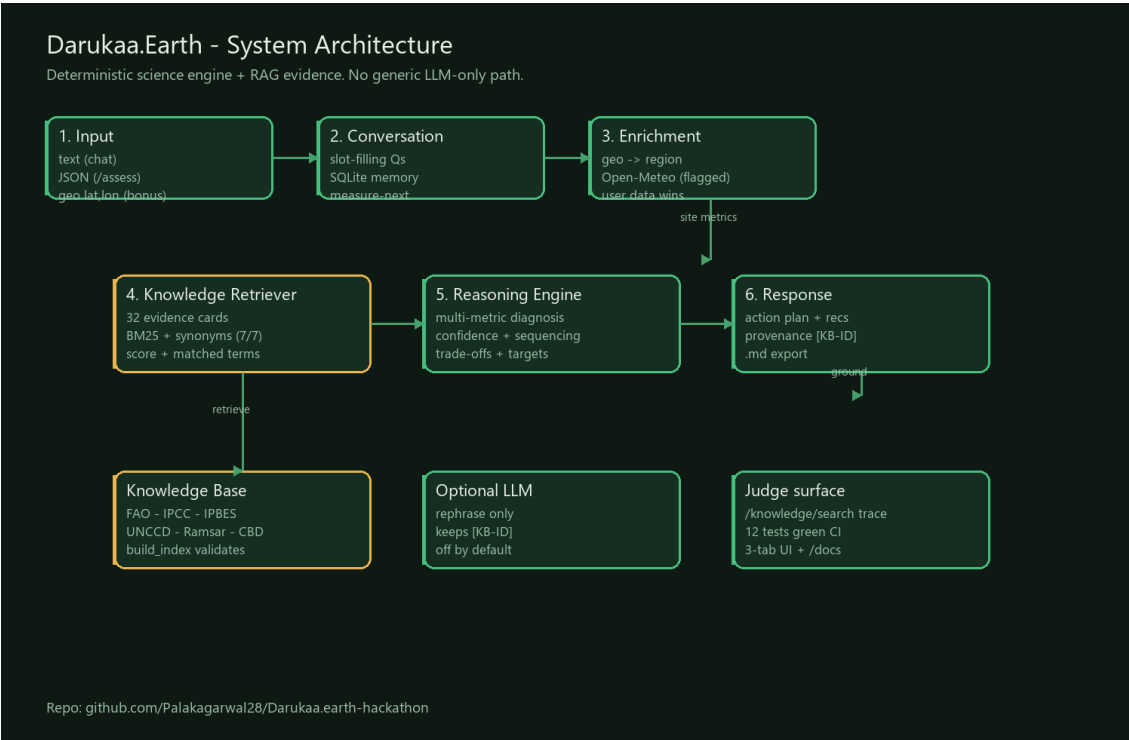


Figure — end-to-end architecture: deterministic engine + RAG evidence.

Stage	Module	Role
1. Input	Chat / Assess	Text, structured JSON, geo lat/lon (bonus)
2. Conversation	src/conversation.py	Metric extraction; clarifying Qs; SQLite memory
3. Enrichment	src/geo.py, integrations.py	Region inference; optional flagged climate fill
4. Retrieval	src/retriever.py, bm25.py	BM25 over 32 cards; kb_id + score + matched terms
5. Reasoning	src/reasoner.py	Diagnosis, confidence, sequencing, trade-offs, targets
6. Response	UI + JSON API	Phased plan, recs, provenance [KB-ID], .md export

4. Knowledge system (critical requirement)

- 32 evidence cards: soil pH/SOC/moisture, land use/cover, species richness/habitat diversity, temperature/rainfall, pollution/deforestation — plus eDNA (KB26) and soil-card (KB27) protocols.
- Sources: FAO, IPCC AR6, IPBES 2016/2019/2023, UNCCD, Ramsar, CBD + peer-reviewed meta-analyses.
- Each card: intervention, mechanism, metrics, effect size, horizon, confidence, source, region fit, conditions, contraindications. Validated by `python -m knowledge.build_index`.

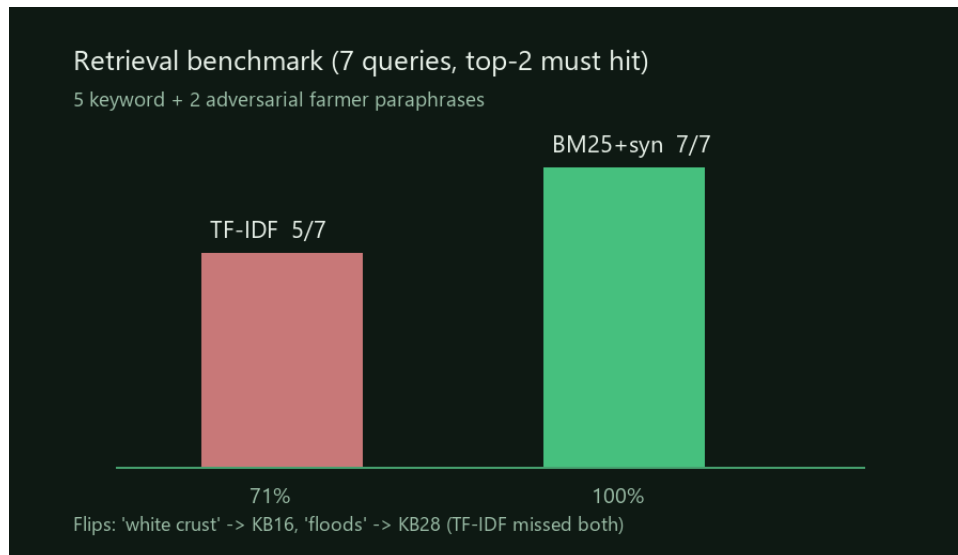
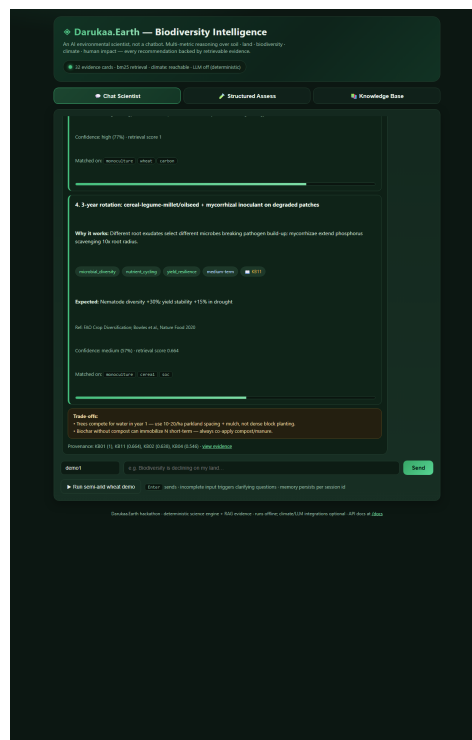


Figure — retrieval benchmark: BM25+synonyms 7/7 vs TF-IDF 5/7.

The two adversarial farmer paraphrases ('white crust on field, salty' → KB16; 'field floods every monsoon' → KB28) miss on TF-IDF and hit on BM25 — the measured case for the default backend. Full method + table: [knowledge/retrieval_benchmark.md](#).

5. Conversational intelligence

- Vague input ('biodiversity is declining') → targeted clarifying questions, never a guess.
- Multi-turn memory per session id; JSON overrides memory; new text merges in.
- *measure_next* ranks missing inputs by decision impact, driving follow-ups.
- Inputs: free text, structured JSON (/assess), geo-coordinates with region inference.



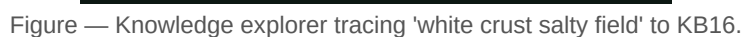
■ **Darukaa.Earth – Biodiversity Intelligence**
We're developing a novel, first-of-its-kind, global biodiversity intelligence platform, connecting the world's biodiversity data, knowledge, and expertise – into a powerful, unified biodiversity intelligence.

■ 12 years on, 100+ national clients, 10M+ of data points



- GET /knowledge/search — scores + matched terms for any query.

- _____



8. API reference

Endpoint	Purpose
POST /chat	Text + memory; clarifying Qs when incomplete
POST /assess	Structured JSON site → full scientific report
GET /knowledge/search	Retrieval trace (scores, matched terms)
GET /provenance?ids=	Exact evidence cards cited
GET /sessions/{sid}	Conversation history
GET /health, /integrations/status, /docs	Status + interactive docs

9. Evaluation summary

Metric	Result	Verify
Retrieval benchmark	BM25 7/7, TF-IDF 5/7	knowledge/retrieval_benchmark.md
Latency (median)	BM25 2.3 ms; assess() 2.7 ms; HTTP 13.4 ms	ARCHITECTURE.md
Tests	12/12 pass	python -m pytest tests -q; CI
Variables combined	6 (wheat demo)	POST /assess variables_count
Offline	Identical, no network	Disconnect + rerun demo

10. Local setup (2 minutes)

- `pip install -r requirements.txt`
- `python -m pytest tests -q`
- `python -m uvicorn api.main:app --host 127.0.0.1 --port 8000` (or double-click `run_local.ps1`)
- Open `http://127.0.0.1:8000` — shareable demos: `?tab=chat&demo=1`, `?tab=assess&demo=1`, `?tab=kb&demo=1&q=...`

GitHub: <https://github.com/Palakagarwal28/Darukaa.earth-hackathon> — no deployment, keys, or credentials required.